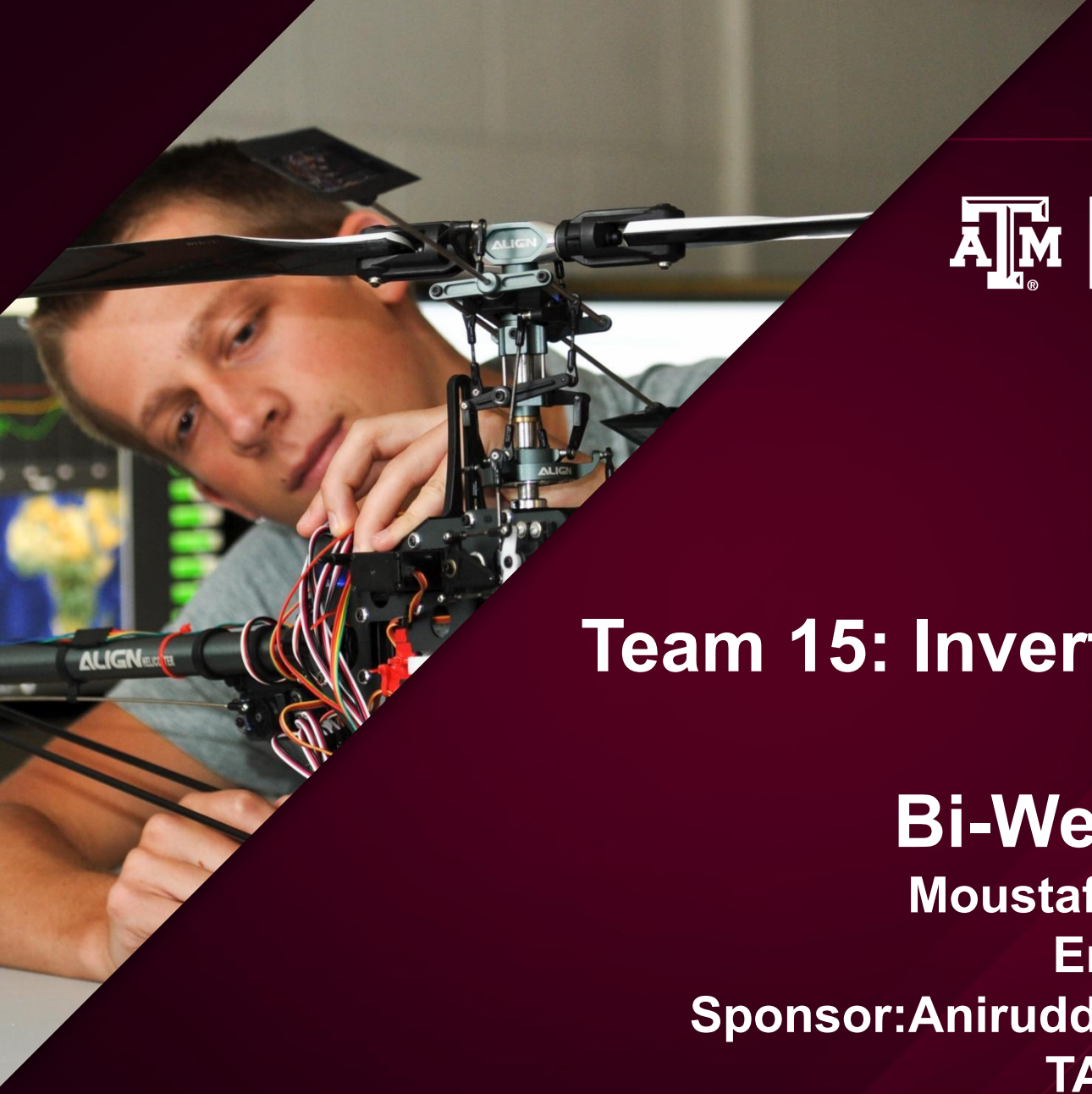


Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

Team 15: Inverted Pendulum Robot 1

Bi-Weekly Update 5

Moustafa Khalil, Kaitlynn Ly
Eric Chang, Gage Fox
Sponsor:Aniruddha Datta, Dr. Lusher
TA: Sabyasachi Gupta

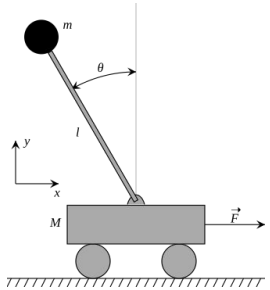
Project Summary

Problem statement:

- Autonomous warehouse robots generally experience instability when handling unbalanced loads or making sudden maneuvers. These movements can cause tilting, making balance a key challenge.

Our Solution Will:

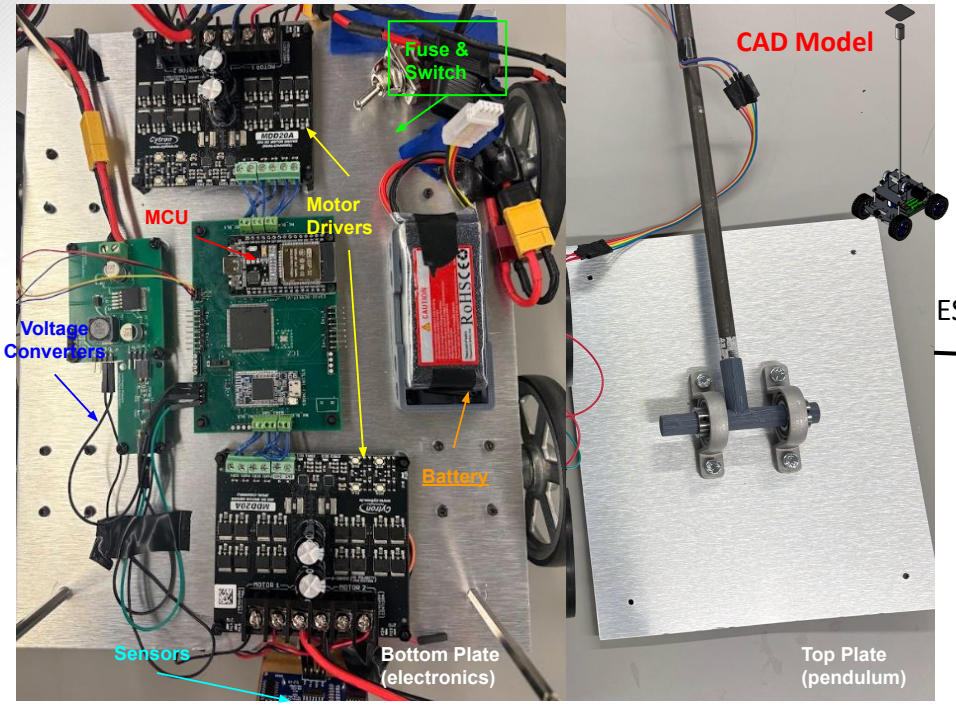
- Deliver an autonomous, self-navigating robot that maintains stability of an unstable payload while moving through constrained spaces.
- Integrate mechanical, sensing, power, and control subsystems into a cohesive, microcontroller-based platform designed for reliability and safety.



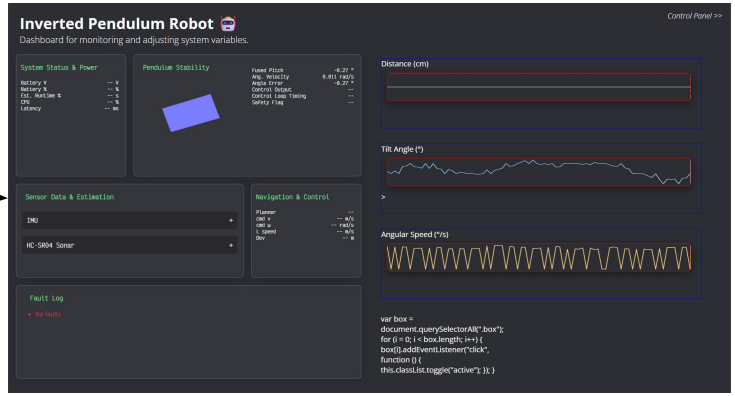
Side View

Isometric View

Integrated System Diagram



ESP-32 comm.



User Interface

Integrated System Architecture

Project Timeline

Subsystem Designs and Testing (complete by 2/9)	Integration of controls, sensors, and MCU (complete by 2/16)	Integration of MCU and motors (to complete by 2/20)	Full System Integration (complete by 3/6)	Full System Testing (complete by 3/20)	System Validation (complete by 4/1)	Demo and Final Report (complete by 4/20)

Controls Algorithm

Moustafa Khalil

Accomplishments since last update 80 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Validated readings from IMU and Ultrasonic sensors • Ongoing validation of inner loop • Tuning robot • Tested inner loop and achieved 3-4 seconds of balancing	• Continue validating inner loop • Continue to find optimal gains

Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

Controls Algorithm

Moustafa Khalil

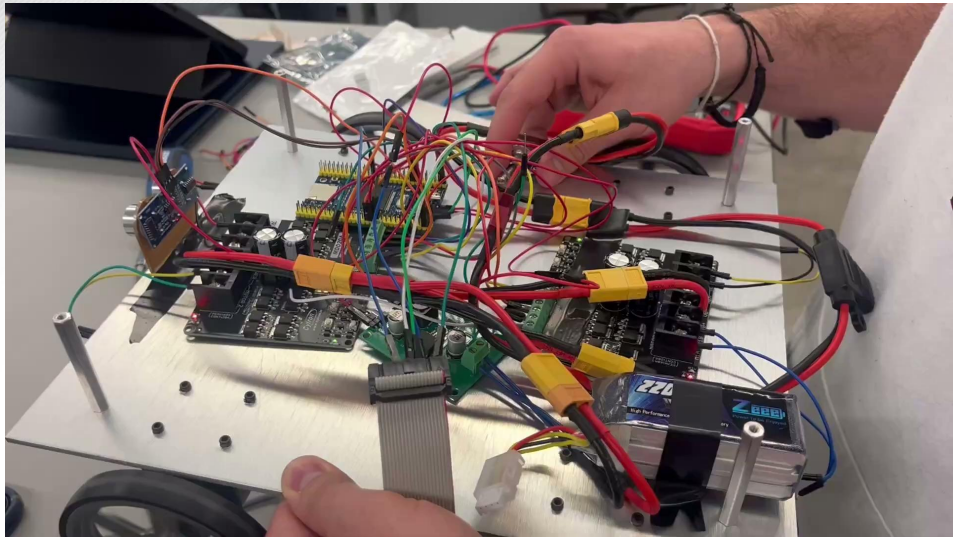
Power & Mechanical

Kaitlynn Ly

Accomplishments since last update 30 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Validated wiring and component stability under motion and vibration • Validated stable power delivery during system startup and initial balancing attempts • Contributing to debugging instability during balancing tests	• Assist in resolving instability during balancing tests

Power & Mechanical

Kaitlynn Ly



Validation of powered ultrasonic sensor supply while streaming data

Validation of Power Delivery During Robot Balancing Attempt

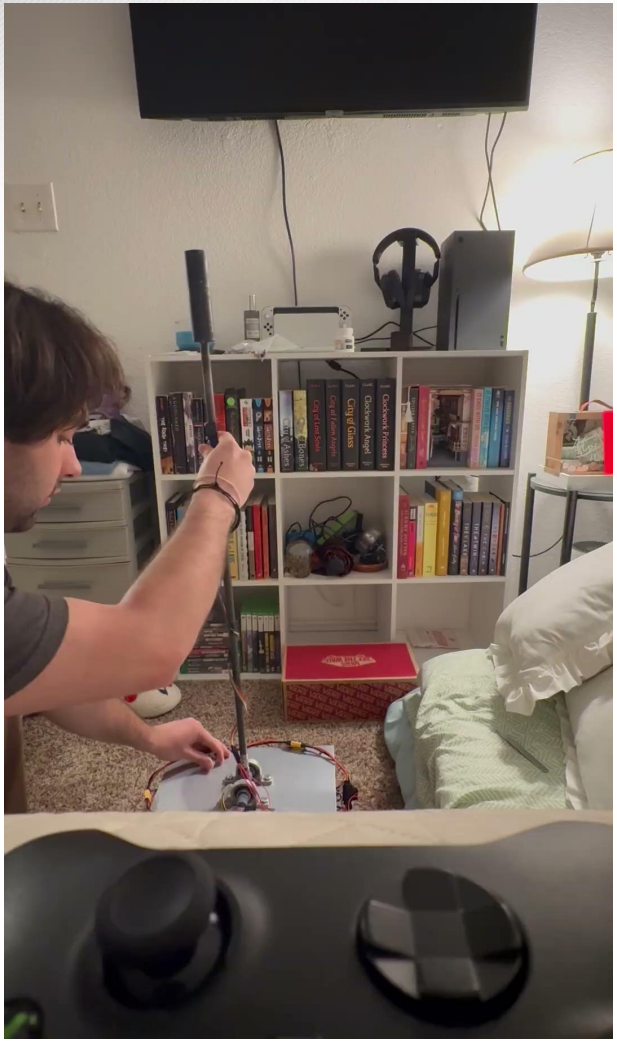
Logic MCU and PCB

Gage Fox

Accomplishments since last update 70 hrs of effort	Ongoing progress/problems and plans until the next presentation
• System reactivity testing on complete robot. • Logic PCB integration • Logic PCB Rev. 3 • Speed Optimizations	• PCB Rev. 3 Integration • Balance Testing • DMA Implementation • Switch to SPI on IMU

Logic MCU and PCB

Gage Fox



Sensors & Navigation

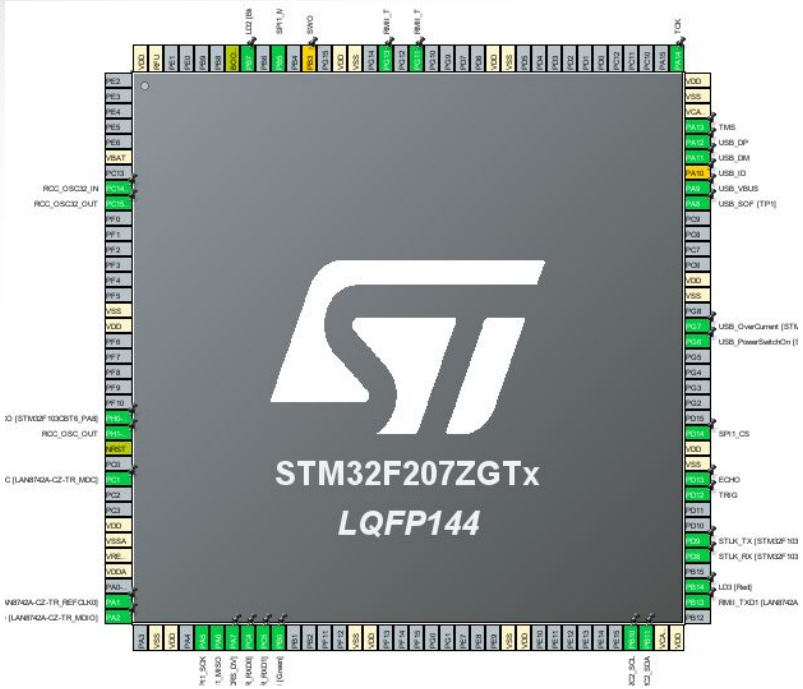
Eric Chang

Accomplishments since last update 36 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Established ESP-STM wireless communication • Validated continuous data transmission between MCU and wireless module • Assessed system responsiveness under different control gain settings	• SPI communication for IMU & validation • Minimize noise throughout the system • Further testing of IMU comms speed with DMA and SPI

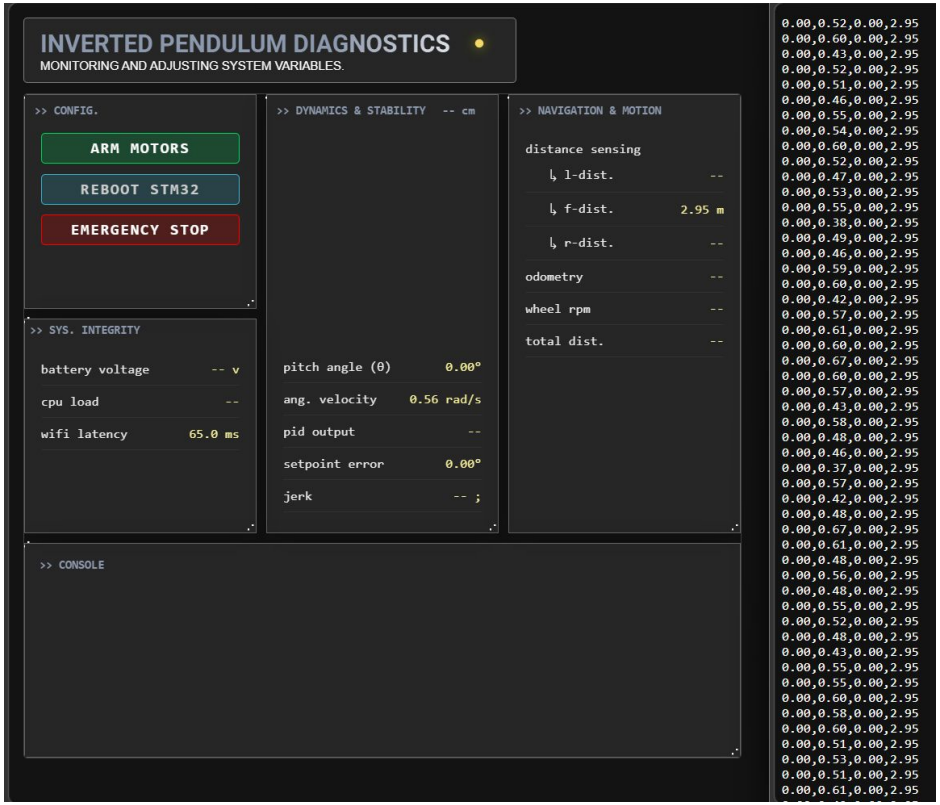
Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

Sensors & Navigation

Eric Chang



Hardware Configuration

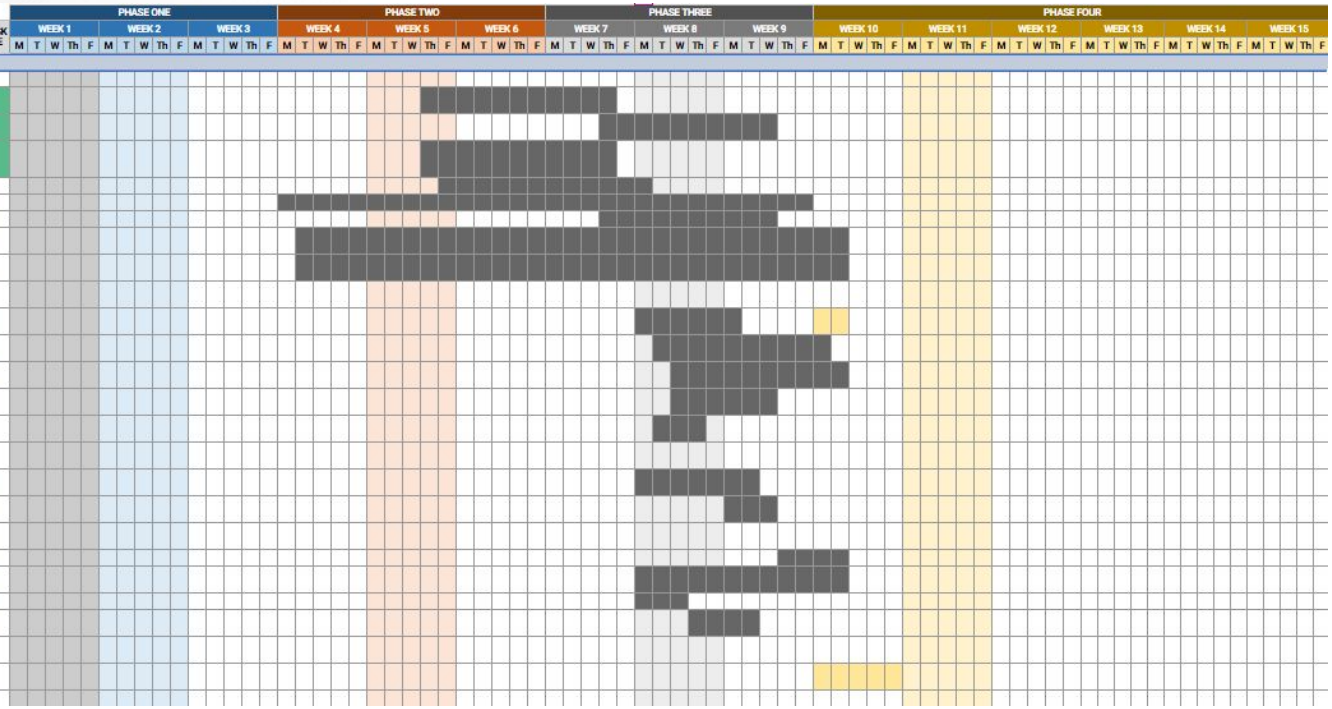


UI for System Monitoring

Dwight Look College of ENGINEERING TEXAS A&M UNIVERSITY

Execution Plan

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION	PCT OF TASK COMPLETE
3	Performance Tuning and System Validation					
3.1	Power and Mechanical Construction	Kaitlynn L.				
3.1.1	Verify regulated 5V/3.3V rails under MCU load	Kaitlynn L.	2/12/26	2/26/26	3	100
3.1.2	Verify power stability during motor start up and braking	Kaitlynn L.	2/26/26	3/9/26	3	100
3.1.3	Provide stable regulated power to IMU and ultrasonic sensors during full system operation	Kaitlynn L.	2/12/26	2/26/26	6	100
3.1.4	System Power Integration	Kaitlynn L.	2/15/26	3/1/26	5	100%
3.1.4	Integration Debug and Rework	Kaitlynn L.	2/1/26	3/15/26	15	100%
3.1.5	Verify full system power on sequence	Kaitlynn L.	2/26/19	3/15/26	16	100%
3.1.6	Electrical Component Mounting & Retention	Kaitlynn L.	2/27/19	3/16/26	10	95%
3.1.7	Final mechanical assembly and structural integration	Kaitlynn L.	2/28/19	3/16/26	12	95%
3.2	Controls subsystem construction	Moustafa K.				
3.2.1	Controls code modified onto MCU	Moustafa K.	02/10/26	3/1/26	15	80%
3.2.2	Controls and sensors integration	Moustafa K.	02/02/26	3/5/26	16	75%
3.2.3	Verify Sensor output is properly read and executed	Moustafa K.	02/10/26	3/10/26	16	50%
3.2.4	Verified control output on physical robot	Moustafa K.	03/19/26	3/26/26	7	0%
3.2.5	Verify controls and mcu integration	Moustafa K.	03/26/26	4/2/26	6	0%
3.2.6	Subsystem Validation	Moustafa K.	04/02/26	4/9/26	7	0%
3.3	Validate sensor performance during dynamic operation	Eric C.	2/26/26	3/5/26	9	70%
3.3.1	Identify changes in bias, drift, and noise under motion	Eric C.	3/5/26	3/8/26	3	40%
3.3.2	Verify encoder accuracy during movement	Eric C.	3/19/26	3/26/26	7	0%
3.4	Finalize logging for integrated system	Eric C.	3/26/26	3/31/26	5	80%
3.5	Tune estimator and controller gains; robustness testing	Eric C.	4/2/26	4/9/26	7	60%
3.6	Tune all communication protocols	Gage F.	3/1/26	3/4/26	3	60%
3.6.1	Fine-tune DMA protocols and peripheral priority	Gage F.	3/5/26	3/10/26	5	30%
3.7	Implement PWM breaking safety features	Gage F.	3/11/26	3/14/26	3	0%
3.8	Determine and Implement noise reductions	Gage F.	3/16/26	3/20/26	5	40%
3.9	Optimize MCU code	Gage F.	3/23/26	3/27/26	3	50%



Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

Validation Plan

Paragraph #	Task	Specifications and Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.1.	Logic Calculation Rate	Validate through timing tests on MCU execution cycle; must complete calculations in < 50ms	Measured using STLINKV2 Debugger	Tested	Gage
3.2.1.3	Data Transfer Rates	Measure message latency with time logs; must be < 100 ms difference	Measured using STLINKV2 Debugger and Analog Discovery Kit	Tested	Gage
3.2.3.1.7	Oscillator Precision	Validate clock stability through frequency tests; ensure deviation < 10ppm		Tested	Gage
3.2.3.1.4	Logic PCB Functionality	Verify that Logic PCB functions as intended without signal interferenece between components	Run test signals through all traces and read the resultant waveforms of all traces using an Analog Discovery Kit	Tested	Gage
3.2.3.1	I2C Communication	Verify that I2C Communication with IMU is functioning without error	Run the IMU alongside the MCU and compare the MCU's readings with the sensor data GUI.	Tested	Gage
3.2.5.1	SPI Master-Slave Communication	Verify proper Master-Slave communication between MCU and Wireless Module	Send and recieve test signals between the MCU (master) and Wireless Module (slave) and output these signals to their respective hosts.	Tested	Gage
3.2.3.1.7	Simultaneous Peripheral Controls and Comms	Validate that all peripherals operate as intended whilst being operated altogether.	Test run that the motors can be controlled with incoming data from all sensors, all at the same time using the MCU.	Tested	Gage
3.2.1.3	Sonar Read Distance	Validate using logged ultrasonic data (CSV) and IMU; confirm ≥95% obstacle detection at distances ≥2 cm and ±3 cm accuracy within 0.2-1.0 m	Use a meter stick to track distances, record N amount of trials at different positions, analyze collected results	Tested	Eric
3.2.1.3	Sensor Field-of-view Testing	Verify that the ultrasonic sensor coverage meets the ±15° design requirements	Use a meter stick to track distances, record N amount of trials at different positions, analyze collected results	Tested	Eric
3.2.3.1.5	IMU steady-state: bias, noise, drift	Verify IMU steady-state metrics: accel bias ±0.02 g, gyro bias ±0.5 °/s, accel noise σ ≤ 0.03–0.05 g, gyro noise σ ≤ 0.3–0.5 °/s, drift ≤1 °/min	Hold the IMU steady (without external disturbances) for an extended period, analyze collected results	Tested	Eric
3.2.5.1	Telemetry/GUI Latency	Mean latency < 200 ms with minimal packet loss	A clock will measure the time delay between the data sent from hardware and data received by the web interface	Tested	Eric
3.2.1.3	IMU and Sonar Dynamic Measurements	Mean error ≤2 ° during motion, mean distance error ≤ 3 cm	Keep the sensors in motion and compare recorded readings against reference values	Untested	Eric
3.2.4	Multi-sensor Coordination	Verify that sensor readings are ≥90% accurate while sensors are active, readings show minimal interference	Run all sensors simultaneously and check for false readings	Tested	Eric
3.2.5.1	Logging Test	Ensure the system is capable of logging all data during a complete run	Output sensor and control data to the web interface during motion and verify that data is accurate and timely	Tested	Eric
3.2.3.1.1	Motor Power Consumption	Measure motor current at multiple voltages under both free-spin and load conditions; confirm total motor power draw is ~40-45 W	Read motor current under different volatges	Tested	Kaitlynn
3.2.3.1	Verify Input Power Delivery	Measure system voltage under load; confirm range is within 11.1-12.6V	Use an oscilloscope to measure voltage under different loads	Tested	Kaitlynn
3.2.3.3	Buck Converter Regulation	Record output voltage under different loads; verify 5V stability	Verify 5V stability under multiple loads using an oscilloscope	Tested	Kaitlynn
3.2.3.4	LDO Validation	Measure regulated output with oscilloscope; confirm 3.3V stability	Measured 3.3V stability	Tested	Kaitlynn
3.2.3.1.2	MCU Power Delivery	Measure 3.3V at MCU during code execution; confirm stable operation	Measure 3.3V at MCU during code execution and confirm stable operation	Tested	Kaitlynn
3.2.3.1.6	Sensor Power Stablity	Measure sensor supply while streaming data; confirm no dropouts	Measure sensor supply while streaming data	Tested	Kaitlynn
3.2.3.5	Motor Driver Power	Measure driver input voltage/current during motor operation	Measure motor current at multiple voltages under both free-spin and load conditions; confirm total motor power draw is ~40-45 W	Tested	Kaitlynn
3.2.3.1	System Power Sequencing	Confirm safe startup; power logic rails first, then motor power	Confirm safe startup; power logic rails first, then motor power	Tested	Kaitlynn
3.2.3.1.7	Control Algorithm Validation	Validate stability across various operating conditions and confirm tuned gains achieve target response time	Verified that the controller was able to successfully balance the inverted pendulum	Tested	Moustafa
3.1.2.1.7	Initial Condition Response (LQR)	In simulation, initialize the pendulum at -10° offset (without PID input) and observe the angle v. time graph as it stabilizes	In simulation, initialize the pendulum at -10° offset (without PID input) and observe the angle v. time graph as it stabilizes	Tested	Moustafa
3.2.3.1.7	Step Response Tracking (PID)	Validate tracking performance by applying a 1.0 m step input; confirm rise time is < 2.5 s, overshoot is < 10%	Validate tracking performance by applying a 1.0 m step input; confirm rise time is < 2.5 s, overshoot is < 10%	Tested	Moustafa
3.2.3.1	Control Algorithm and MCU Integration	Able to run with the desired rise time, and give commands to rest of robot	Deploy compiled control loops to the MCU; verify PWM signal generation and command transmission frequency meet timing requirements for the desired rise time.	Tested	Moustafa
3.2.3.1.5	Control System and Sensor Integration	Able to cleanly read the data from sensors	Perform real-time polling of IMU and encoder registries; verify the control loop processes filtered data without lag or noise-induced jitter.	Untested	Moustafa
3.2.3.1.1	Controls System and Power Integration	Control subsystem successfully gives correct power to the robot to deal with various obstacles	Subject the robot to physical barriers; monitor current spikes and duty cycle adjustments to ensure the power system sustains high-torque demands.	Untested	Moustafa
3.2.1.1	Upright Pendulum Stability	Robot successfully balances pendulum in the upright position for more than one minute with both control loops running	Place robot upright, enable both control loops, and measure duration of stable balance	Untested	All
3.2.1.2	Inner Loop Stability	Robot successfully balancing pendulum with only the inner loop running for more than 30 seconds	Place robot upright, disable the outer control loop with the innner loop enabled. Measure the duration of stable balance	Untested	All
3.2.5.1	Over Tilt Failure Propagration	If tilt exceeds 15 degrees, motors are immediately disabled; system enters safe state and requires reset to enable	Intentionally tilt robot past 15 degrees and verify motor and safe state behavior	Untested	All
3.2.1.2	Maze Navigation Performance	Robot successfully navigates a maze from start to finish; maintains upright stability through navigation	Test navigation across multiple maze configurations to stimulate unknown demo conditions; record compelteion success and stability over repeated trials	Untested	All
4.1.1	Remote Command Execution	The system shall successfully execute remote commands transmitted from the user interface; defined as achieving >95% execution accuracy on 20 trials and execution latency of <200ms	Commands will be sent from the web interface to the MCU by ESP32 over 20 trials. The response will be recorded and latency will be measured as time from transmission to response received	Untested	All

Validation Plan

3.2.1.1	Upright Pendulum Stability	Robot successfully balances pendulum in the upright position for more than one minute with both control loops running	Place robot upright, enable both control loops, and measure duration of stable balance	Untested	All
3.2.1.2	Inner Loop Stability	Robot successfully balancing pendulum with only the inner loop running for more than 30 seconds	Place robot upright, disable the outer control loop with the innner loop enabled. Measure the duration of stable balance	Untested	All
3.2.5.1	Over Tilt Failure Propagration	If tilt exceeds 15 degrees, motors are immediately disabled; system enters safe state and requires reset to enable	Intentionally tilt robot past 15 degrees and verify motor and safe state behavior	Untested	All
3.2.1.2	Maze Navigation Performance	Robot successfully navigates a maze (traverse tight corners) from start to finish; maintains upright stability through navigation	Test navigation across multiple maze configurations to stimulate unknown demo conditions; record completion success and stablilty over repeated trials	Untested	All
4.1.1	Remote Command Execution	The system shall successfully execute remote commands transmitted from the user interface; defined as achieving >95% execution accuracy on 20 trials and execution latency of <200ms	Commands will be sent from the web interface to the MCU by ESP32 over 20 trials. The response will be recorded and latency will be measured as time from transmission to response received.	Untested	All

Thank You!

Inverted Pendulum Robot 1

Any Questions?